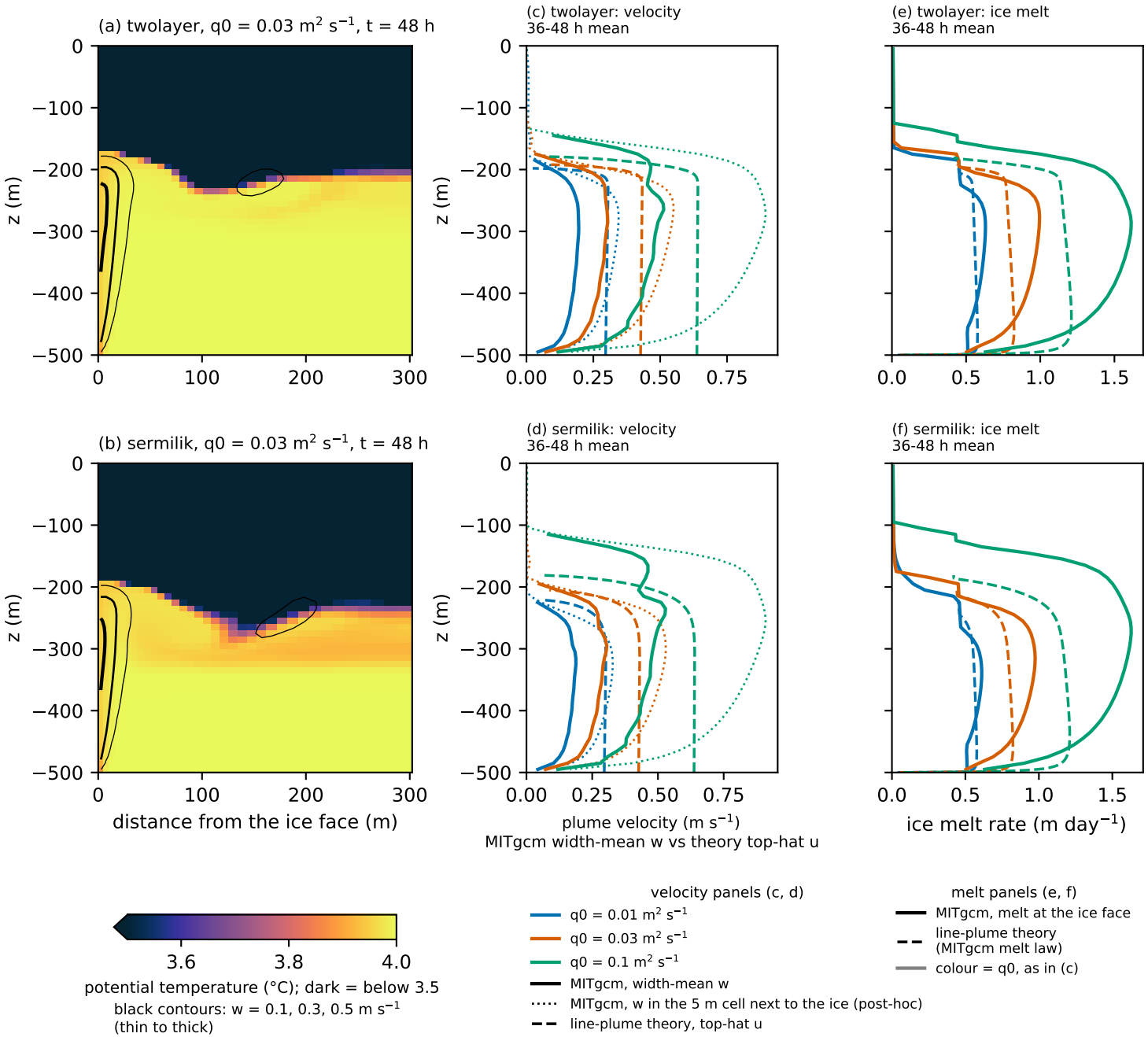

(a) twolayer, q0 = 0.03 m^2 s^{-1}, t = 48 h
(b) sermilik, q0 = 0.03 m^2 s^{-1}, t = 48 h
z (m)
distance from the ice face (m)
(c) twolayer: velocity
36-48 h mean
(d) sermilik: velocity
36-48 h mean
plume velocity (m s^{-1})
MITgcm width-mean w vs theory top-hat u
(e) twolayer: ice melt
36-48 h mean
(f) sermilik: ice melt
36-48 h mean
ice melt rate (m day^{-1})
potential temperature (°C); dark = below 3.5
black contours: w = 0.1, 0.3, 0.5 m s^{-1}
(thin to thick)
velocity panels (c, d)
q0 = 0.01 m^2 s^{-1}
q0 = 0.03 m^2 s^{-1}
q0 = 0.1 m^2 s^{-1}
MITgcm, width-mean w
MITgcm, w in the 5 m cell next to the ice (post-hoc)
line-plume theory, top-hat u
melt panels (e, f)
MITgcm, melt at the ice face
line-plume theory
(MITgcm melt law)
colour = q0, as in (c)